

INTERWAR RACING PLANES

DESIGNERS OF INTERWAR RACING PLANES concentrated exclusively on speed over short distances at low altitude. Concerns such as range or higher altitude performance were left to the designers of serious commercial and military aircraft. What was needed was an engine delivering maximum horsepower for its weight, allied to a lightweight airframe with minimum drag. Racing planes provided a testbed for high-octane fuels and high-performance engines, as well as for developments in streamlining. But racing remained a sport – with the usual mix of spectacular thrills and competition for international prestige. Throughout this period, seaplanes were often the fastest aircraft, because of the almost unlimited takeoff run from open water, which made up for the drag associated with floats. Until 1931, the Schneider Trophy seaplane competition provided the focus for attempts on the world speed record, with competitors funded by their governments. Landplane racing flourished in the United States from the late 1920s, eliciting inspired designs from racing pilots like James Wedell or smaller planemakers like the Granville brothers.

SUPER SPORTSTER

A Gee Bee R-1 being flown by Jimmy Doolittle during the 1932 Cleveland Air Races. Doolittle's R-1 set a landplane world speed record of 294mph (473kph).



Caudron C.460



The C.460 was the sensation of the 1936 US National Air Race season. Designed by Marcel Riffard at the long-established Caudron company, it set an airspeed record of 314mph (506kph) in 1934. At the 1936 Thompson Trophy, it trounced the American opposition.

ENGINE 370hp 8-litre Renault Bengali 6-cylinder inline

WINGSPAN 22ft 1in (6.7m) **LENGTH** 23ft 3in (7.1m)

TOP SPEED 314mph (505kph) **CREW** 1

PASSENGERS None

Curtiss R3C-2 Racer

For the 1925 season, the Curtiss company (which had built one of the earliest seaplanes in 1911) produced the R3C-1 and the R3C-2 (with floats). On October 25, Jimmy Doolittle flew the R3C-2 to a new world speed record of 246mph (395kph).



ENGINE 565hp Curtiss V-1400 12-cylinder liquid-cooled

WINGSPAN 22ft (6.7m) **LENGTH** 22ft (6.7m)

TOP SPEED 246mph (395kph) **CREW** 1

PASSENGERS None

Granville Model R-1 Super Sportster



The most extraordinary of all the early 1930s racing designs were the highly distinctive Gee Bee racers. In 1932, the Granville Brothers produced the ultimate racer – the Super Sportster – which came in two versions: the R-1 and the longer range 525hp R-2.

ENGINE 745hp supercharged P&W R-1340 Wasp 9-cylinder radial

WINGSPAN 25ft (7.6m) **LENGTH** 17ft 9in (5.4m)

TOP SPEED 294mph (473kph) **CREW** 1

PASSENGERS None

Hughes H-1 Racer

Conceived by the wealthy but eccentric Howard Hughes, the H-1 was designed to be the world's fastest landplane. That aim was achieved on September 13, 1935, when Hughes flew the H-1 at 352.2mph (566.7kph).



ENGINE 700hp supercharged Pratt & Whitney Twin Wasp radial

WINGSPAN 31ft 9in (9.6m) **LENGTH** 27ft (8.2m)

TOP SPEED 352mph (567kph) **CREW** 1

PASSENGERS None

Macchi M.39

In November 1926, the Italians won the Schneider Trophy for fastest seaplane at Hampton Roads, Norfolk, Virginia, with their remarkable M.39 monoplane racer. Designed by Mario Castoldi, its winning speed over the closed circuit was 246.5mph (396.6kph). The Americans had won the previous two years with converted Curtiss fighter planes and were hoping for a record third victory, so that they could keep the trophy. In the event, they were thwarted, and the US Navy withdrew from the Schneider Trophy competition altogether. On November 17, the winning pilot, Mario de Bernardi, subsequently flew the M.39 to a new world air speed record of 258.9mph (416.6kph).

ENGINE 800hp Fiat AS-2 V12 liquid-cooled

WINGSPAN 30ft 6in (9.3m) **LENGTH** 22ft 1in (6.7m)

TOP SPEED 259mph (417kph) **CREW** 1

PASSENGERS None

